

The Use of AI in Kinematics

Hands-on tutorial for validation-centered human-AI solver development

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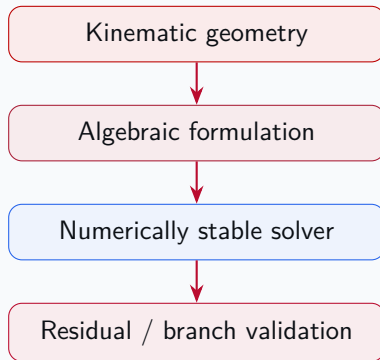
45-minute tutorial + 15-minute Q/A. Based on two accepted ASME IDETC/CIE 2026 papers; examples trace to original kinematics literature and companion repositories.

Why kinematics is a hard AI domain

The difficulty is not syntax

- ▶ Geometry of motion
- ▶ Branches and singularities
- ▶ Polynomial elimination
- ▶ Numerical stability
- ▶ Independent validation

Tutorial foundation: accepted ASME IDETC/CIE 2026
Paper Nos. DETC2026-193961 and DETC2026-193726.



What AI agents add

Contribution

- ▶ Faster implementation loop
- ▶ Broader tool orchestration
- ▶ Stricter testable workflow

chat

IDE

CAS

MCP

Prompt

role, source, problem, expected checks



Code

Python, Mathematica, C++, web app



Run

terminal, CAS, test scripts, reports



Fix

inspect residuals, branches, method choice



The agent is

useful only when the loop includes execution and checks.

Two ASME IDETC 2026 papers behind this tutorial

Vision paper

Where we are in the path towards AGI in kinematics?

- ▶ 5-level intelligence hierarchy
- ▶ AGI benchmark framing
- ▶ General 6R IK as a stress test

Accepted ASME IDETC/CIE 2026 Paper No. DETC2026-193961

Practical guide

Solving Kinematics Problems by Leveraging the Power of Large Language Models

- ▶ Environment and prompt structure
- ▶ Planar/spherical linkage examples
- ▶ Spatial SS synthesis and Stewart–Gough FK

Accepted ASME IDETC/CIE 2026 Paper No. DETC2026-193726

Publication information

Hai-Jun Su and J. Michael McCarthy. Proceedings of the ASME 2026 International Design Engineering Technical Conferences and Computers and Information in Engineering Conference (IDETC/CIE 2026), Houston, TX, August 23–26, 2026.

Source: Paper metadata in the two ASME IDETC/CIE 2026 manuscripts.

Public repositories behind the papers

Paper 1 repositories

- ▶ [IK_6R_RR_1993](#): prompt, reference PDFs, Mathematica derivation, Python/C++ 6R IK solvers, benchmark reports.
- ▶ [llm_kinematics_deep_research_report](#): open-problem PDF reports and cleaned deep-research prompts.

Paper 2 repositories

- ▶ [gdl_llm_reproduction](#): GDL prompts, scripts, outputs, and reports.
- ▶ [llm_ss_synthesis_liao_2001](#): SS prompt, Mathematica/Python solver, validation report.
- ▶ [llm_FK_Stewart_Husty_1996](#): Husty prompt, Wolfram script, output, test report.

generated code posted

prompts posted

reports posted

validation artifacts posted

The tutorial examples point to public data, generated code, prompts, and reports in these repositories.

Kinematics repositories on haijunsu-osu: ecosystem view

Direct paper companions

IK_6R_RR_1993, GDL, SS dyad, Stewart FK, deep research.

Teaching and interactive apps

kinematics_ug, UR5e_Teach_Play, UR5e robot, robot demo.

Adjacent AI-assisted IK

llm_ik_6r_wrist, ik_6r_parallel, kinematics_eq_solvers.

Classical and compliant mechanisms

CBCM, PRB 3R, rocker synthesis, spherical 4R.

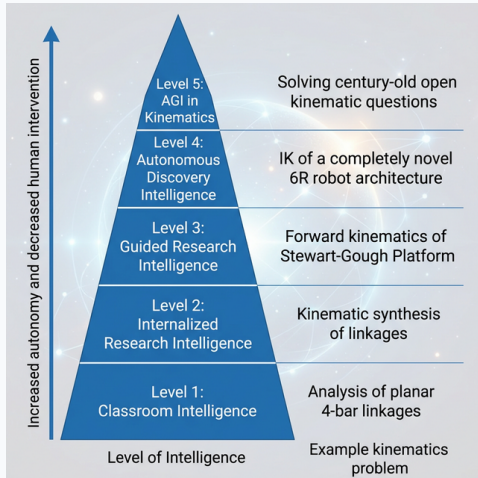
41 public repos scanned

30+ kinematics / mechanisms / robotics related

full list in [github_kinematics_repo_scan.md](#)

Source: public GitHub API profile scan of haijunsu-osu on May 12, 2026.

A five-level map for AI in kinematics



L	Mode	Example
1	Classroom	Planar four-bar analysis
2	Internalized	Exact dyad synthesis
3	Guided research	Stewart–Gough FK or 6R IK with paper
4	Discovery	Novel solvable architectures
5	AGI	General 6R IK from scratch

Boundary claim

Current systems are effective at Level 3 and show Level 4 potential only with structured expert guidance.

Source: Paper 1: accepted ASME IDETC/CIE 2026, Paper No. DETC2026-193961, hierarchy and discussion sections.

Roadmap for the next 45 minutes

1. Introduction

Where AI is useful, and where it is not



2. Environment + live demo

ChatGPT, VS Code, Codex, Claude Code



3. Kinematics examples

Four-bar to general 6R IK



4. Future and education

Responsible use and student learning

Working rule

Treat AI as a laboratory assistant that must be given source material, constraints, tools, and tests.

do not trust plausible math

verify every branch

Environment: four levels of tool integration

1. Conversational chat

Brainstorm, explain, outline, critique



2. IDE-based code generation

Write files, run scripts, inspect output



3. Computational tool integration

Python, NumPy/SciPy, Mathematica, MATLAB



4. MCP / sandbox self-debugging

Tool access plus bounded execution

Tutorial posture

Use chat for thinking. Use agent mode for files and tests. Use CAS when the algebra is the problem.

Source: Paper 2: accepted ASME IDETC/CIE 2026, Paper No. DETC2026-193726, AI Agent Environment Setup.

Minimum setup for the live demo

Interactive tools

- ▶ Online ChatGPT mode
- ▶ VS Code workspace
- ▶ Codex extension / app
- ▶ Claude Code as another agent interface
- ▶ GitHub repository or local folder

Execution tools

- ▶ Python 3.11
- ▶ NumPy and SciPy
- ▶ Mathematica / wolframscript when needed
- ▶ Browser for web-app demo
- ▶ Terminal for tests and reports

[source files](#)

[agent mode](#)

[run tests](#)

[commit artifacts](#)

Source: Paper 2: accepted ASME IDETC/CIE 2026, Paper No. DETC2026-193726, minimum requirements and workflow discussion.

Hands-on demo time: online ChatGPT mode

Ask

Explain the kinematics task and likely solution path



Probe

What information is missing? What proof would verify it?



Convert

Turn the discussion into a structured agent prompt

Demo message

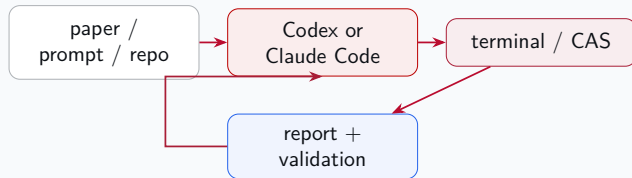
Chat is useful for planning and critique, but it is not evidence that code works.

no filesystem

no local tests

fast framing

Hands-on demo time: VS Code + LLM extensions



What to show live

- ▶ Open folder
- ▶ Attach source context
- ▶ Ask for files
- ▶ Run command
- ▶ Read failure
- ▶ Iterate

Hands-on demo time: two examples

Demo A: web app

- ▶ Prompt: make a small kinematics visualizer
- ▶ Example: four-bar linkage or IK branch explorer
- ▶ Run locally in browser
- ▶ Use visuals to teach geometry

Demo B: paper to code

- ▶ Give paper/source file
- ▶ Ask for solver implementation
- ▶ Require validation harness
- ▶ Run and inspect residuals

known solution count

planted root

residual check

FK- IK -FK

Prompt checklist before asking the agent

Context: role, source files, notation

Task: exact solver or app behavior

Requirements: branches, tolerances, edge cases

Deliverables: files, commands, report

Validation: residuals, roots, known examples

Source control: preserve artifacts

Closing rule

The prompt is not complete until the pass/fail evidence is specified.

Source: Paper 2: accepted ASME
IDETC/CIE 2026, Paper No.
DETC2026-193726, Prompt Structure.

Case studies across the complexity ladder



Original sources: McCarthy and Soh, *Geometric Design of Linkages*, Springer, 2010; Liao and McCarthy, "Seven-position synthesis of a spatial SS chain," *J. Mechanical Design*, 2001; Husty, "An algorithm for solving the direct kinematics of general Stewart-Gough platforms," *Mechanism and Machine Theory*, 1996; Raghavan and Roth, "Kinematic analysis of general six-revolute manipulators using dialytic elimination," *IJRR*, 1993.

Example 1: planar four-bar analysis

Chapter 2 setup

$$A = (a \cos \theta, a \sin \theta)$$

$$B = (g + b \cos \psi, b \sin \psi)$$

$$\|B - A\|^2 - h^2 = 0$$

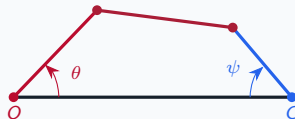
Scalar output equation

$$A_\theta \cos \psi + B_\theta \sin \psi = C_\theta$$

$$A_\theta = 2ab \cos \theta - 2gb$$

$$B_\theta = 2ab \sin \theta$$

$$C_\theta = g^2 + b^2 + a^2 - h^2 - 2ag \cos \theta$$



2 assembly branches

loop residual

Original source: McCarthy and Soh, *Geometric Design of Linkages*, Springer, 2010, Ch. 2.

Example 1 details: what the agent must preserve

(2.46) Link dimensions: $a = |AO|$, $b = |BC|$, $g = |CO|$, $h = |BA|$



(2.47) Fixed coupler length: $\|B - A\|^2 - h^2 = 0$



(2.50)–(2.52) Solve $\psi = \text{atan2}(B_\theta, A_\theta) \pm \arccos(C_\theta / \sqrt{A_\theta^2 + B_\theta^2})$



(2.55)–(2.57) Recover ϕ from loop equations using the same branch



Harness Reject non-assembly cases; verify scalar, loop, and length residuals

Numerical guardrails

- ▶ Use `atan2`, not a one-quadrant arctangent.
- ▶ Check $A_\theta^2 + B_\theta^2 - C_\theta^2 \geq 0$ before `acos`.
- ▶ Label branches by whether B lies above or below line AC .
- ▶ Treat folded/near-folded cases as special evidence, not failures.

Local run evidence

- ▶ Test links: $a = 1$, $b = 2$, $g = 2.5$, $h = 2$.
- ▶ Angles: 20° , 40° , 80° , 120° , 160° .
- ▶ Two valid branches at each angle.
- ▶ Max residual: $< 5 \times 10^{-15}$; status PASS.

Source: McCarthy and Soh, *Geometric Design of Linkages*, Springer, 2010, Ch. 2, Secs. 2.3.1–2.3.2 and 2.4.4–2.4.5. Local artifact: `McCarthy_GDL/prompts_GDL/ch2/Ch2_4R_solver.py`.

Example 2: planar RR/PR dyad synthesis

Chapter 5 RR dyad

$$(G - P_{1i}) \cdot [A(\phi_{1i}) - I](W_1 - P_{1i}) = 0$$

- ▶ Unknowns: fixed pivot $G = (x, y)$ and moving pivot $W_1 = (\lambda, \mu)$.
- ▶ Bilinear structure: linear if either G or W_1 is selected.
- ▶ Five precision positions produce finite RR candidates.

Eq. 5.17

Burmester curves

Chapter 5 PR dyad

$$|S, [A(\phi_{1i}) - I](W_1 - P_{1i})| = 0, \quad S = (1, m)^T$$

- ▶ Unknowns: guide slope m and moving pivot W_1 .
- ▶ Four precision positions determine a PR chain.
- ▶ Same agent pattern: derive, enumerate, rank-check, validate.

Local artifact evidence

RR: 4 real solutions, residuals $\leq 3.2 \times 10^{-15}$; PR: 1 real solution, Eq. 5.66 residual 2.2×10^{-16} .

Original source: McCarthy and Soh, *Geometric Design of Linkages*, Springer, 2010, Ch. 5. Local artifact: McCarthy_GDL/prompts_GDL/ch5/report.md.

Example 2: spherical RR dyad synthesis

Chapter 9 design constraint

$$G^T(A_{1i} - I)W_1 = 0$$

- ▶ G is the fixed axis; W_1 is the moving axis.
- ▶ Same dyad idea, now on the unit sphere.
- ▶ Validation is axis-aware: v and $-v$ describe the same line.

Eq. 9.25 constraints



4 positions: center/circling-axis cubic cones



5 positions: degree-6 elimination polynomial

Original source: McCarthy and Soh, *Geometric Design of Linkages*, Springer, 2010, Ch. 9. Local artifacts: `ch9_rr_4pos_report.md`; `ch9_rr_5pos_report.md`.

Four-position workflow

- ▶ Center-axis cone: Eqs. 9.45–9.46.
- ▶ Circling-axis cone: Eqs. 9.53–9.54.
- ▶ Local run exported both cone equations and 30 validated dyad samples.

Five-position workflow

- ▶ Eqs. 9.55–9.59 give $P(x) = \det(D)$.
- ▶ Local run: degree 6, six simple roots, two validated real dyads.
- ▶ Residuals: 10^{-49} scale for 4-position samples; 10^{-58} – 10^{-60} for validated 5-position dyads.

Example 3: seven-position spatial SS dyad

SS chain constraint

$$\|A_i p + d_i - B\|^2 = r^2, \quad i = 1, \dots, 7$$

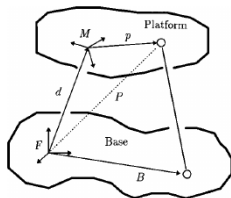


Fig. 2 An SS chain with fixed pivot $B = (B_x, B_y, B_z)^T$ and moving pivot $P = (P_x, P_y, P_z)^T$ in the fixed frame F

Source/figure: Liao and McCarthy, "Seven-position synthesis of a spatial SS chain,"

J. Mechanical Design, 2001, Fig. 2.

What is specified

- ▶ Fixed pivot B
- ▶ Moving pivot p
- ▶ Seven spatial poses
- ▶ Constant-distance constraint

7 poses



linear equations in p



15×15 matrix polynomial



roots in B_z

SS dyad: the important debugging lesson

Failed route

$$\det(\mathcal{M}(B_z)) = 0$$

- ▶ Large determinant expansion
- ▶ Catastrophic cancellation
- ▶ Plausible code, wrong route

Expert-guided route

$$\mathcal{M}(B_z) = M_0 + M_1 B_z + \cdots + M_4 B_z^4$$

- ▶ Companion matrix pencil
- ▶ SVD back-substitution
- ▶ Validate against known solutions
- ▶ 20 real solutions; residuals near 10^{-13}

Original source: Liao and McCarthy, "Seven-position synthesis of a spatial SS chain," *J. Mechanical Design*, 2001.

Example 4: Stewart–Gough forward kinematics

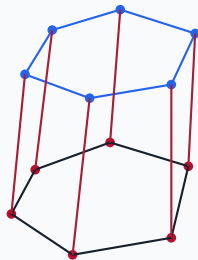
Problem

Given six leg lengths between base and platform points, find all assembly configurations.

$$\mathcal{P}_{40}(p) = 0$$

- ▶ Study parameter mapping
- ▶ Resultant cascade
- ▶ Up to 40 configurations

Original sources: Husty, "An algorithm for solving the direct kinematics of general Stewart–Gough platforms," *Mechanism and Machine Theory*, 1996; Raghavan, *J. Mechanical Design*, 1993.



Stewart–Gough: why the agent succeeded

Study quadric and leg constraints



Follow Husty's elimination order



Use exact rational arithmetic



Extract degree-40 factor



Verify planted root $p = -1$

Lesson

High algebraic degree was manageable because the published algorithm decomposed into explicit CAS operations.

40

polynomial degree

$p = -1$

known planted root

Original source: Husty, “An algorithm for solving the direct kinematics of general Stewart–Gough platforms,” *Mechanism and Machine Theory*, 1996; companion repo validation report.

Example 5: general 6R inverse kinematics

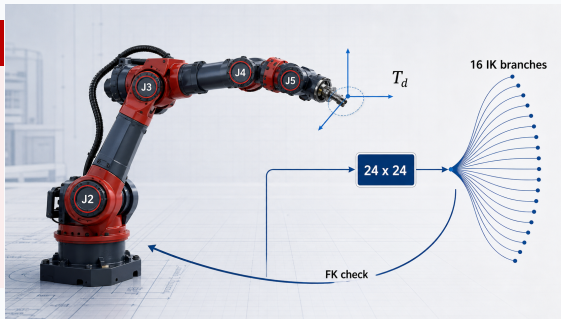
Raghavan–Roth pipeline

- ▶ Form 14 scalar equations
- ▶ Eliminate (θ_1, θ_2)
- ▶ Dyalitic lifting
- ▶ Left null-space projection
- ▶ Solve characteristic polynomial

24 × 24 pencil

16 branches

FK residual check



Original source: Raghavan and Roth, “Kinematic analysis of general six-revolute manipulators using dialytic elimination,” *IJRR*, 1993. Supporting numerical formulation: Manocha and Canny, *IEEE T-RA*, 1994; Manseur and Doty, *IJRR*, 1989. Generated figure shows the 6R chain, branch fan, and FK validation loop.

General 6R IK: benchmark facts and numerical robustness

16

Manseur–Doty real IK solutions

14

Raghavan–Roth scalar equations

24×24

eigenvalue pencil

Critical expert intervention

The original 6R IK paper does not specify the numerically robust final root solve. The companion-matrix generalized eigenvalue formulation had to be supplied through expert guidance.

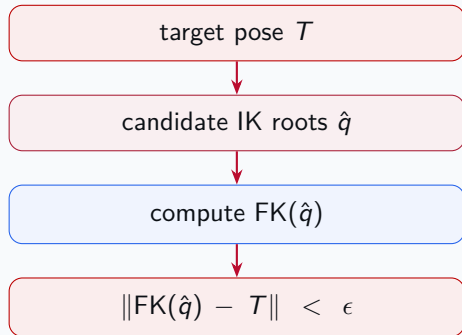
sign/index errors

wrong monomial basis

unstable direct roots

Original sources: Raghavan and Roth, “Kinematic analysis of general six-revolute manipulators using dialytic elimination,” *IJRR*, 1993; Manocha and Canny, *IEEE T-RA*, 1994; Manseur and Doty, *IJRR*, 1989. Companion repo benchmark report.

FK–IK–FK is the cheap verifier

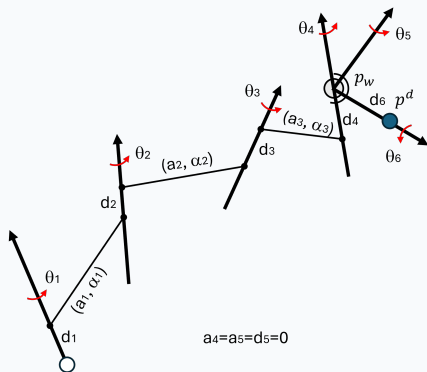


What it catches

- ▶ Hallucinated roots
- ▶ Branch omissions
- ▶ Numerical instability
- ▶ Bad conventions

Original source: Raghavan and Roth, “Kinematic analysis of general six-revolute manipulators using dialytic elimination,” *IJRR*, 1993; validation against Manseur and Doty, 1989.

Example 6: 6R spherical-wrist IK



Architecture and solver route

- ▶ Intersecting wrist: $a_4 = 0$, $d_5 = 0$, $a_5 = 0$; all other DHM parameters remain arbitrary.
- ▶ Wrist center reduces position IK to a spatial 3R problem.
- ▶ Orientation reduces to a spherical wrist subproblem.
- ▶ Helper solvers enumerate all valid trigonometric branches.

723 robots

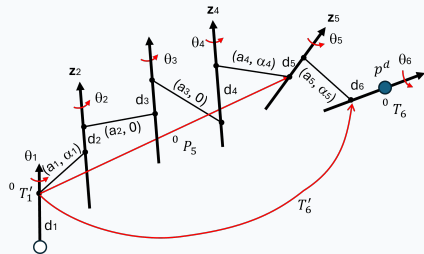
100% success

1.22 ms

Published source: Su, *Mechanism and Machine Theory*, 222:106392, 2026.

Repo: `llm_ik_6r_wrist`. Helper solvers: `kinematics_eq_solvers`.

Example 7: 6R parallel-joint IK



Architecture and solver route

- ▶ Parallel axes: $\alpha_2 = 0$, $\alpha_3 = 0$; all other DHM parameters remain arbitrary.
- ▶ Solve (q_1, q_6) through a bilinear two-angle system.
- ▶ Recover q_5 from the orientation constraint.
- ▶ Finish (q_2, q_3, q_4) as a planar 3R IK problem.

119 robots

100% success

1.79 ms

Published source: Su, *Mechanism and Machine Theory*, 222:106392, 2026.

Repo: [ik_6r_parallel](#). Helper solvers: [kinematics_eq_solvers](#).

Autonomy changes with problem structure

Case	Agent help	Human role	Verifier
Four-bar / dyads	direct code	select source, inspect	residuals, branches
SS dyad	candidate solver	choose matrix pencil	20 sols., planted root
Stewart–Gough	CAS translation	preserve elimination order	degree 40, $p = -1$
General 6R IK	solver coding	guide route/root solve	16 roots, FK–IK–FK
Spherical-wrist 6R	prompt-to-solver build	specify decoupling	723 robots, FK checks
Parallel-joint 6R	bilinear + planar code	encode geometry cue	119 robots, stress tests

Practical lesson

The hard part is not asking for code. The hard part is specifying the stable mathematical route and the evidence that proves it worked.

Original example sources: McCarthy and Soh, *Geometric Design of Linkages*, Springer, 2010; Liao and McCarthy, “Seven-position synthesis of a spatial SS chain,” *J. Mechanical Design*, 2001; Husty, “An algorithm for solving the direct kinematics of general Stewart–Gough platforms,” *Mechanism and Machine Theory*, 1996; Raghavan and Roth, “Kinematic analysis of general six-revolute manipulators using dialytic elimination,” *IJRR*, 1993; Su, *Mechanism and Machine Theory*, 222:106392, 2026.

Case-study takeaway

Good tasks for agents

- ▶ Translate clear algorithms
- ▶ Build test harnesses
- ▶ Generate reports
- ▶ Compare residuals

Do not outsource blindly

- ▶ Elimination order
- ▶ Branch completeness
- ▶ Numerical method choice
- ▶ Interpretation of failures

source material

structured prompt

executable solver

independent validation

Original example sources plus companion repository reports for generated code and validation artifacts.

Current boundary: useful, not autonomous AGI



Boundary claim

Current LLMs reproduce complex documented pipelines, but still need expert guidance for novel symbolic routes.

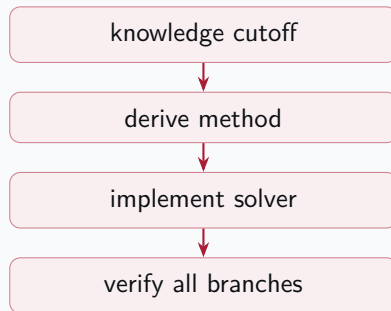
Source: Paper 1: accepted ASME IDETC/CIE 2026, Paper No. DETC2026-193961, current AI intelligence level and limitations.

What would stronger AGI evidence look like?

A stronger test

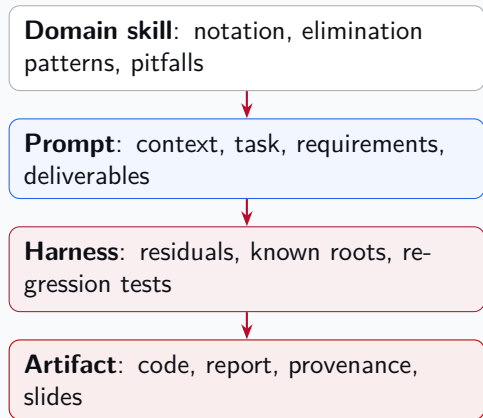
Remove post-1980 knowledge, then ask the AI to independently derive the general 6R IK solution or an equivalent spatial 7R closed-loop framework.

- ▶ Choose the symbolic strategy
- ▶ Select stable numerical methods
- ▶ Recover all valid branches
- ▶ Pass independent validation



Source: Paper 1: accepted ASME IDETC/CIE 2026, Paper No. DETC2026-193961, AGI benchmark framing.

Future workflow: domain skills plus validation harnesses



Why this matters

Reusable skill documents can encode expert algebraic guidance and reduce repeated prompting burden.

Source: Paper 1: accepted ASME IDETC/CIE 2026, Paper No. DETC2026-193961, future work; Paper 2: accepted ASME IDETC/CIE 2026, Paper No. DETC2026-193726, workflow template.

Responsible use and disclosure

Human responsibility

- ▶ Select source materials
- ▶ Specify validation criteria
- ▶ Diagnose mathematical failures
- ▶ Review generated content
- ▶ Disclose AI assistance

Do not claim

- ▶ Vendor benchmark without controlled tests
- ▶ Autonomous discovery without evidence
- ▶ Correctness without independent checks
- ▶ Complete branch coverage without proof

Source: Paper 2: accepted ASME IDETC/CIE 2026, Paper No. DETC2026-193726, generative AI declaration and limitations.

Teaching apps and study assistants

Four-bar explorer: drag links, see branches

Dyad synthesis sandbox: task poses to pivots

IK branch visualizer: multiple solutions

NotebookLM / study assistant: summarize, quiz, compare

Upside

Visualization and interaction can make abstract kinematic constraints concrete.

Risk

Students may skip derivation, debugging, and retention if AI gives finished answers too early.

Pedagogy: use AI without hollowing out learning

Require	Forbid as evidence
▶ Hand derivation first ▶ Explain branch structure ▶ Agent-aided implementation ▶ Independent validation report ▶ Reflection on failures	▶ Unrun code ▶ Screenshots without source ▶ Claims without residuals ▶ Copy-pasted derivations ▶ Missing AI-use disclosure

derive

implement

validate

explain

Closing vision

AI as a kinematics laboratory assistant

The best near-term use is not replacing mathematical judgment. It is compressing the implementation loop while making validation more explicit.

- ▶ Researchers formulate better problems.
- ▶ Agents generate and test more code.
- ▶ Students explore more mechanisms.
- ▶ Everyone must verify more carefully.

45 min

tutorial

15 min

Q/A

1 rule

trust evidence, not polish

Backup: papers and public repositories

Accepted ASME IDETC/CIE 2026 papers

- ▶ Su and McCarthy, "Where we are in the path towards AGI in kinematics?" Proceedings of the ASME IDETC/CIE 2026, Houston, TX, Aug. 23–26, 2026, Paper No. DETC2026-193961.
- ▶ Su and McCarthy, "Solving Kinematics Problems by Leveraging the Power of Large Language Models." Proceedings of the ASME IDETC/CIE 2026, Houston, TX, Aug. 23–26, 2026, Paper No. DETC2026-193726.

Mechanism and Machine Theory paper

- ▶ Su, "Large Language Model Assisted Human–AI Collaborative Development of Analytical Inverse Kinematics Solvers for Robot Manipulators," *Mechanism and Machine Theory*, Vol. 222, Article 106392, 2026, doi:10.1016/j.mechmachtheory.2026.106392.

Public artifact repositories

- ▶ `IK_6R_RR_1993`; `llm_kinematics_deep_research_report`
- ▶ `gdl_llm_reproduction`; `llm_ss_synthesis_liao_2001`; `llm_FK_Stewart_Husty_1996`
- ▶ `kinematics_eq_solvers`; `llm_ik_6r_wrist`; `ik_6r_parallel`; `RobotKinematicsCatalogue`

Repos host prompts, generated code/scripts, outputs, validation reports, and source/reference material for Q/A or demo recovery.